ARTIFICIAL INTELLIGENCE

PROJECT 1

Rule-Based AI Chatbot

Industrial Training Kit | Batch 2026
Powered by DecodeLabs

PROJECT REPORT

1. Introduction

Project 1 is the foundation phase described in the DecodeLabs training material. It focuses on control flow, decision-making logic and basic AI concepts through a Rule-Based AI Chatbot.

2. Project Goal

Create a simple rule-based chatbot that responds to predefined user inputs.

3. Key Requirements

Requirement	Implementation
Handle greetings and exit commands	Recognize greeting words and exit commands.
Use if-else logic for responses	Use conditional statements to select responses.
Run in a continuous loop	Keep the conversation running until an exit command is entered.

4. Key Skills

Control flow, decision-making logic and basic AI concepts.

5. Tools and Technology

Python, Jupyter Notebook / Anaconda, if-else statements, while loop and string input handling.

6. Chatbot Implementation

The following implementation directly addresses the project requirements from the DecodeLabs material.

```
print("Welcome to the Rule-Based AI Chatbot!")
print("Type 'bye' or 'exit' to end the chat.")

while True:
    user_input = input("You: ").lower().strip()

    if user_input in ["hello", "hi", "hey"]:
        print("Bot: Hello! How can I help you?")

    elif user_input in ["how are you", "how are you?"]:
        print("Bot: I am fine. Thank you for asking!")

    elif user_input in ["what is your name", "what's your name"]:
        print("Bot: I am a Rule-Based AI Chatbot.")

    elif user_input in ["thank you", "thanks"]:
        print("Bot: You're welcome!")

    elif user_input in ["bye", "exit", "quit"]:
        print("Bot: Goodbye! Have a nice day!")
        break

    else:
        print("Bot: Sorry, I don't understand that input.")
```

7. How the Chatbot Works

1. Display a welcome message.
2. Take user input.
3. Convert the input to lowercase and remove extra spaces.
4. Check the input using if-else conditions.
5. Display the matching response.
6. Continue through the while loop.
7. Stop when the user enters an exit command.

8. Sample Output

```
Welcome to the Rule-Based AI Chatbot!
Type 'bye' or 'exit' to end the chat.

You: hello
Bot: Hello! How can I help you?

You: how are you
Bot: I am fine. Thank you for asking!

You: what is your name
Bot: I am a Rule-Based AI Chatbot.

You: bye
Bot: Goodbye! Have a nice day!
```

[INSERT SCREENSHOT 1 HERE – CHATBOT CODE]

Figure 1: Rule-Based AI Chatbot Code

[INSERT SCREENSHOT 2 HERE – RUNNING CHATBOT OUTPUT]

Figure 2: Chatbot Interaction and Output

9. Testing

User Input	Expected Response
hello / hi / hey	Greeting response
how are you	Status response
what is your name	Chatbot identity response
thank you / thanks	Welcome response
bye / exit / quit	Goodbye response and loop termination
Other input	Unknown-input response

10. Project Flow

Start → Welcome Message → User Input → if-else Decision → Response → Continue Loop → Exit Command → Stop

11. Conclusion

The project demonstrates a basic Rule-Based AI Chatbot using explicit if-else instructions and a continuous loop. It satisfies the stated requirements for greetings, exit commands and decision-making logic. The training material also encourages expanding vocabulary, adding nested conditions and creating a unique chatbot personality.

12. Screenshot Checklist

Add screenshots of the chatbot code and running output. A useful final submission can show a greeting interaction, another predefined response and the exit command.